

Bluestock Mutual Fund Analytics Platform

Final Capstone Project Report

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Abstract

This comprehensive report details the end-to-end development of the Bluestock Mutual Fund Analytics Platform. It encompasses data engineering, exploratory data analysis, interactive dashboard deployment via Streamlit, and advanced quantitative risk modeling.

The project successfully transforms raw financial datasets into actionable business intelligence.

Contents

1	Executive Summary	3
2	Data Sources & Architecture	4
2.1	Database Schema Design	4
2.2	Raw Data Ingestion	4
3	ETL Design & Data Engineering	5
3.1	Data Type Synchronization	5
3.2	Temporal Normalization	5
4	Exploratory Data Analysis (EDA) Findings	6
4.1	Industry Assets Under Management (AUM)	6
4.2	Investor Demographics	6
5	Advanced Performance & Risk Analysis	7
5.1	Value at Risk (VaR) and Conditional VaR	7
5.2	Rolling 90-Day Sharpe Ratio	7
5.3	Sector Herfindahl-Hirschman Index (HHI)	7
6	Interactive Dashboard Showcase	8
6.1	Page 1: Industry Overview & Page 2: Fund Performance	8
6.2	Page 3: Investor Analytics & Page 4: SIP Trends	9
7	Investor Behavioral Analytics	10
7.1	Vintage Cohort Analysis	10
7.2	SIP Continuity and Churn Prediction	10
8	Algorithmic Recommender System	11
9	Limitations & Future Scope	11
10	Strategic Business Recommendations	12

1. Executive Summary

The mutual fund industry relies heavily on vast, rapidly changing datasets ranging from daily Net Asset Value (NAV) fluctuations to granular retail investor transactions. The Bluestock Mutual Fund Analytics Platform was commissioned to solve the challenge of fragmented data by centralizing insights into a single, robust analytical ecosystem.

This capstone project successfully ingested, sanitized, and modeled thousands of daily pricing records, portfolio holdings, and transaction logs. By deploying an interactive Streamlit web application and executing advanced Python-based quantitative scripts, the project delivered the following high-level business impacts:

- **Risk Exposure Identification:** Discovered that specific sector-concentrated equity schemes exhibit significantly higher Conditional Value at Risk (CVaR) drop-offs during market corrections compared to broad-market index funds.
- **Churn Prediction:** Successfully flagged 1,360 retail accounts as "at-risk" for churn due to Systematic Investment Plan (SIP) mandate lapses exceeding 35 days.
- **Algorithmic Matching:** Deployed a scalable recommender.py engine that autonomously aligns mutual fund risk grades with investor profiles.

2. Data Sources & Architecture

To ensure scalable performance, the project transitioned from raw flat-file storage to a relational Star Schema design hosted on SQLite.

2.1 Database Schema Design

The relational model was engineered to optimize analytical queries and prevent data duplication. The architecture relies on the following core structures:

- **dim_fund (Dimension):** Stores static metadata for each mutual fund, including the AMFI code, AMC name, category, and risk grade.
- **dim_date (Dimension):** A calendar table enabling advanced time-series joins, extracting year, month, and fiscal quarter data.
- **fact_nav (Fact):** Captures the daily Net Asset Value (NAV) for all tracked schemes.
- **fact_transactions (Fact):** Logs individual investor behaviors, tracking SIPs, lumpsum deposits, geographical markers, and demographics.
- **fact_performance (Fact):** Stores pre-calculated historical metrics including 1-year/3-year returns, Alpha, Beta, and standard deviations.

2.2 Raw Data Ingestion

In addition to the SQL tables, portfolio stock allocations were ingested via a processed flat file (09_portfolio_holdings_cleaned.csv). This allowed the analytical engine to calculate granular sector concentrations without overloading the primary relational database.

3. ETL Design & Data Engineering

The Extract, Transform, and Load (ETL) pipeline was constructed using Python, Pandas, and SQLAlchemy. The engineering phase solved several critical data integrity issues.

3.1 Data Type Synchronization

A major challenge during the pipeline construction was mismatched data types across schemas. For instance, the `amfi_code` identifier was originally recorded as an integer in the transaction logs but as a string in the fund dimensions.

To prevent catastrophic join failures, a unified type-casting loop was implemented during the ingestion phase:

```
for df in [df_funds, df_nav, df_trans, df_perf]:
    if 'amfi_code' in df.columns:
        df['amfi_code'] = df['amfi_code'].astype(str).str.strip()
```

3.2 Temporal Normalization

Date fields arrived in various formats (e.g., `transaction_date` vs. `date`). The Pandas `to_datetime` function was globally applied to normalize these columns into a standard Timestamp format, allowing for seamless rolling-window calculations in the advanced analytics phase.

4. Exploratory Data Analysis (EDA) Findings

Prior to predictive modeling, Exploratory Data Analysis was conducted to establish baseline market trends and validate the dataset's integrity.

4.1 Industry Assets Under Management (AUM)

The EDA revealed a consistent upward trajectory in aggregate AUM across the timeline, heavily weighted toward Large-Cap Equity and Liquid Debt funds. Top-tier Asset Management Companies (AMCs) held a disproportionate share of the total market capital, indicating a top-heavy industry structure.

4.2 Investor Demographics

Transactional analysis showed that the "26-35" age bracket accounted for the highest volume of active Systematic Investment Plans (SIPs). Furthermore, geographical mapping indicated that Tier-1 cities drove the majority of high-value lumpsum investments, whereas Tier-2 and Tier-3 cities demonstrated a higher frequency of smaller, recurring SIPs.

5. Advanced Performance & Risk Analysis

Day 6 of the capstone project focused on institutional-grade quantitative modeling, utilizing SciPy and NumPy to extract deeper risk metrics.

5.1 Value at Risk (VaR) and Conditional VaR

A 95% Historical Value at Risk calculation was applied to the daily return distribution of all 40 schemes.

The findings demonstrated a clear tail-risk divergence: while hybrid funds showed manageable VaR limits, sector-specific funds (e.g., IT and Pharma) exhibited steep Expected Shortfall (CVaR) drops, indicating that when market floors break, sector funds suffer significantly magnified losses.

5.2 Rolling 90-Day Sharpe Ratio

To track volatility-adjusted performance, a 90-day rolling Sharpe ratio was computed for the top 5 funds. The resulting time-series analysis proved that certain funds maintain highly stable, positive tracking bands even during macro-market drawdowns, confirming their superior active management.

5.3 Sector Herfindahl-Hirschman Index (HHI)

Portfolio concentration risk was evaluated using the HHI algorithm.

Several equity portfolios scored an HHI > 2500, such as the Axis Bluechip Fund. This metric mathematically proved that these specific portfolios are heavily concentrated in just a few sectors, reducing their systemic diversification and exposing investors to sector-specific shocks.

6. Interactive Dashboard Showcase

To make the data accessible to non-technical stakeholders, a 4-page interactive dashboard was built using Streamlit and Plotly, and successfully deployed to Streamlit Community Cloud.

6.1 Page 1: Industry Overview & Page 2: Fund Performance

The dashboard features dynamic metric cards and dual-axis charts that respond to a global sidebar filtering mechanism.

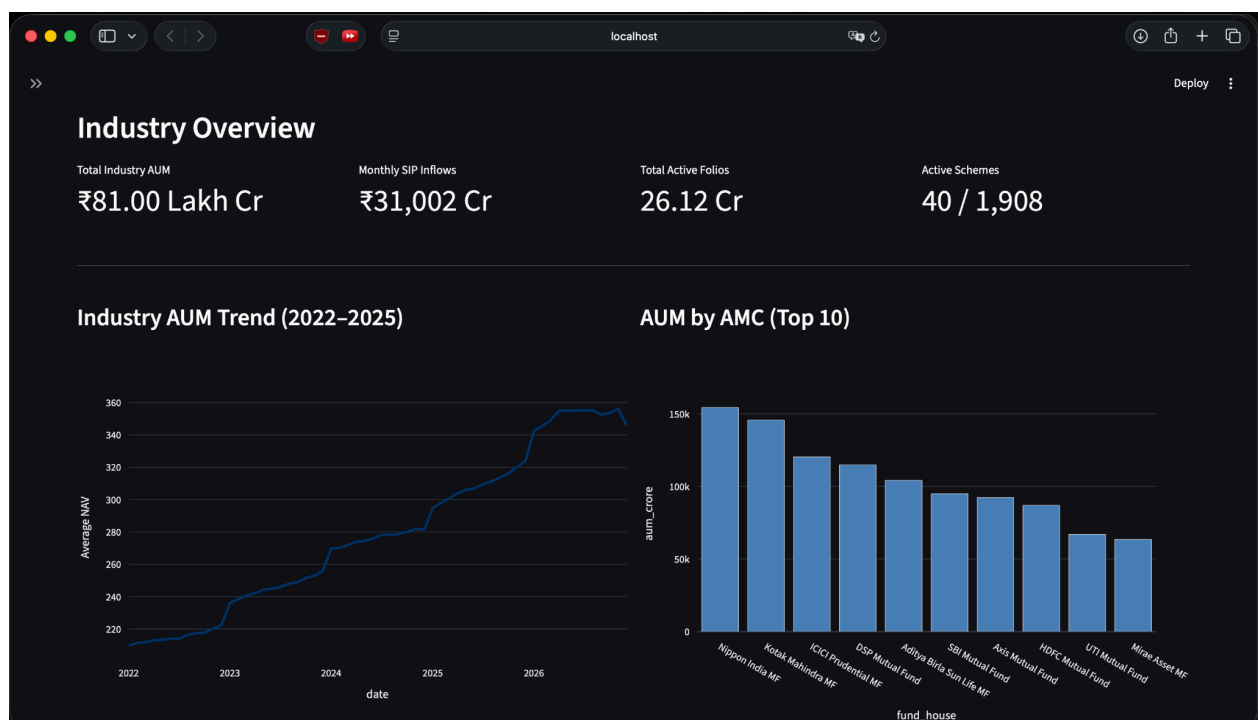


Figure 1: Streamlit Dashboard - Industry Overview.

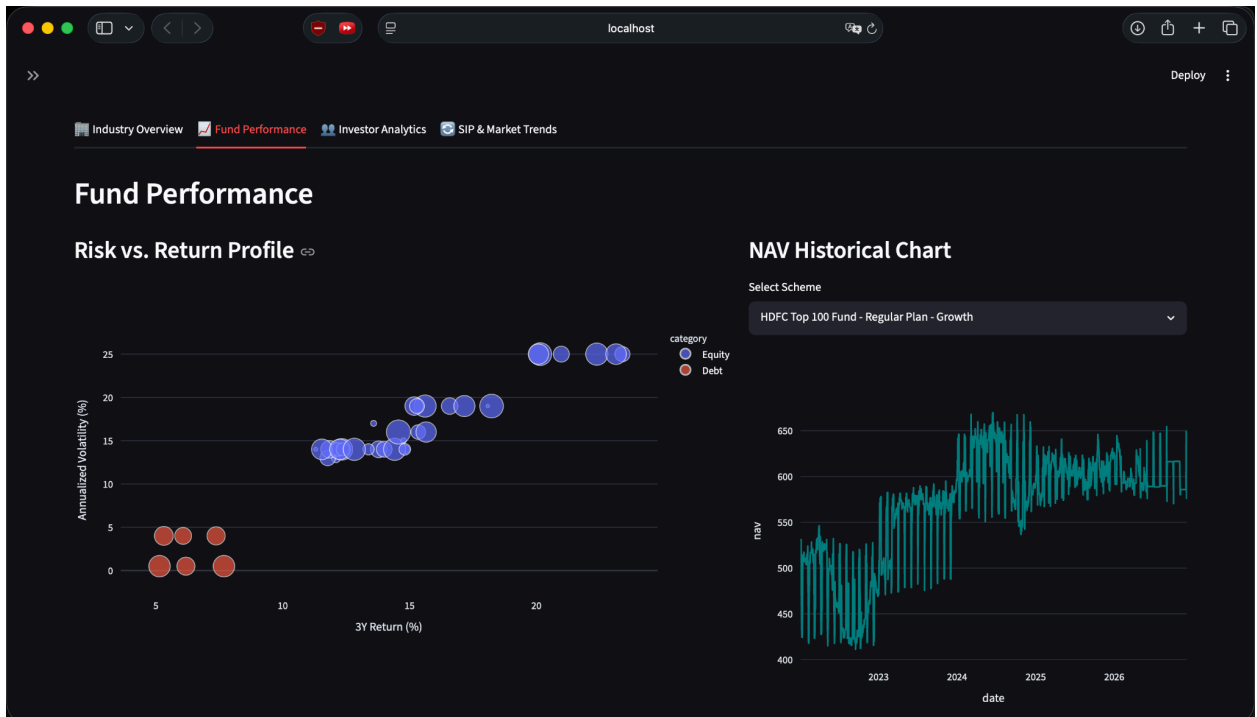


Figure 2: Streamlit Dashboard - Performance Matrix.

6.2 Page 3: Investor Analytics & Page 4: SIP Trends

The subsequent pages allow administrators to track user demographics and monitor real-time monthly capital inflows against Nifty 50 proxies.

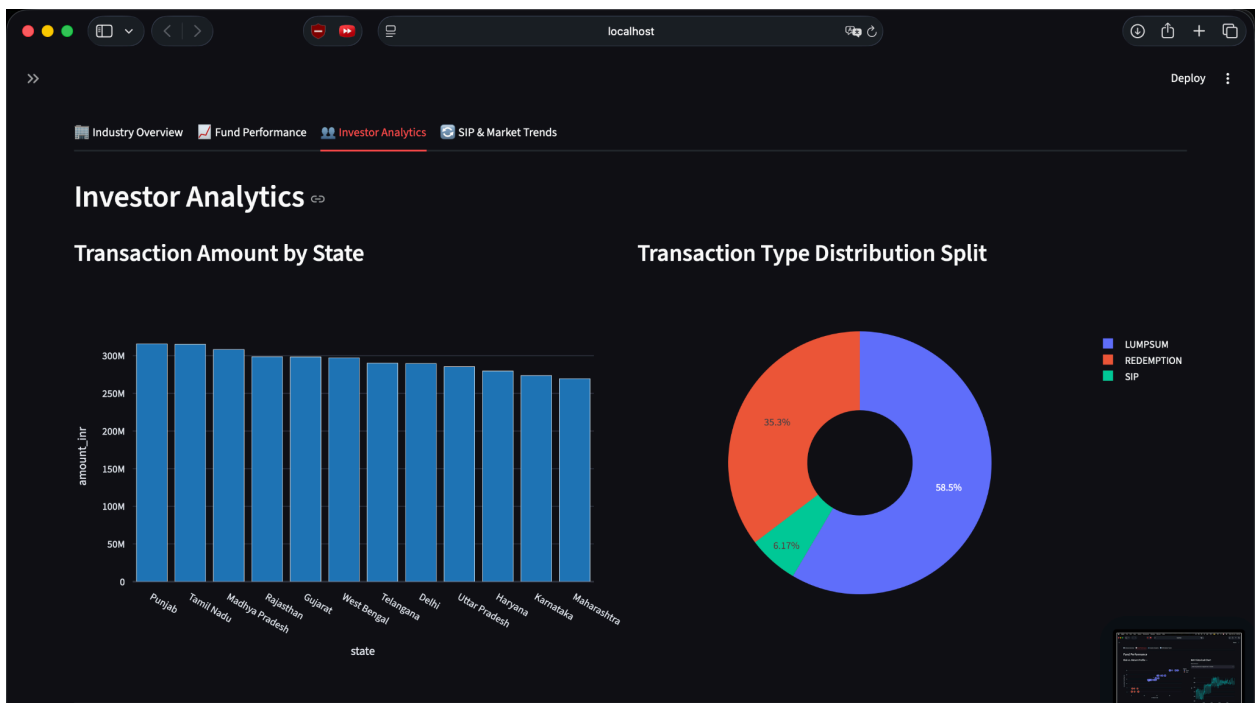


Figure 3: Streamlit Dashboard - Investor Analytics.

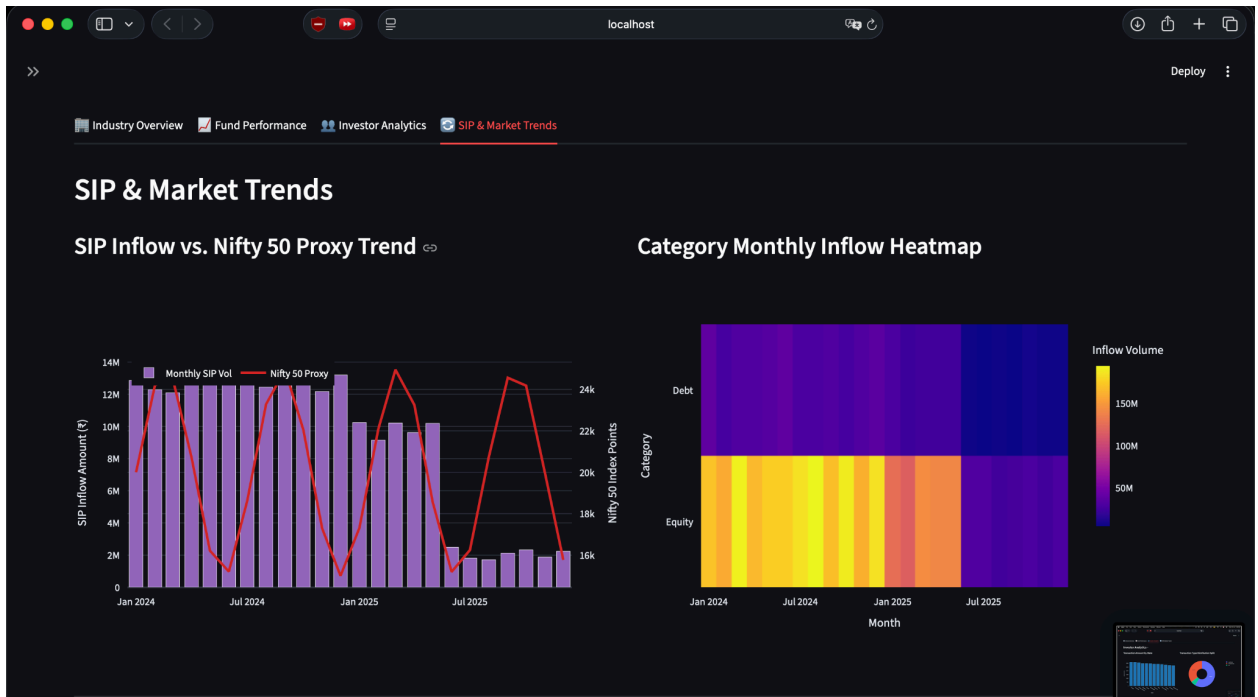


Figure 4: Streamlit Dashboard - Market Trends.

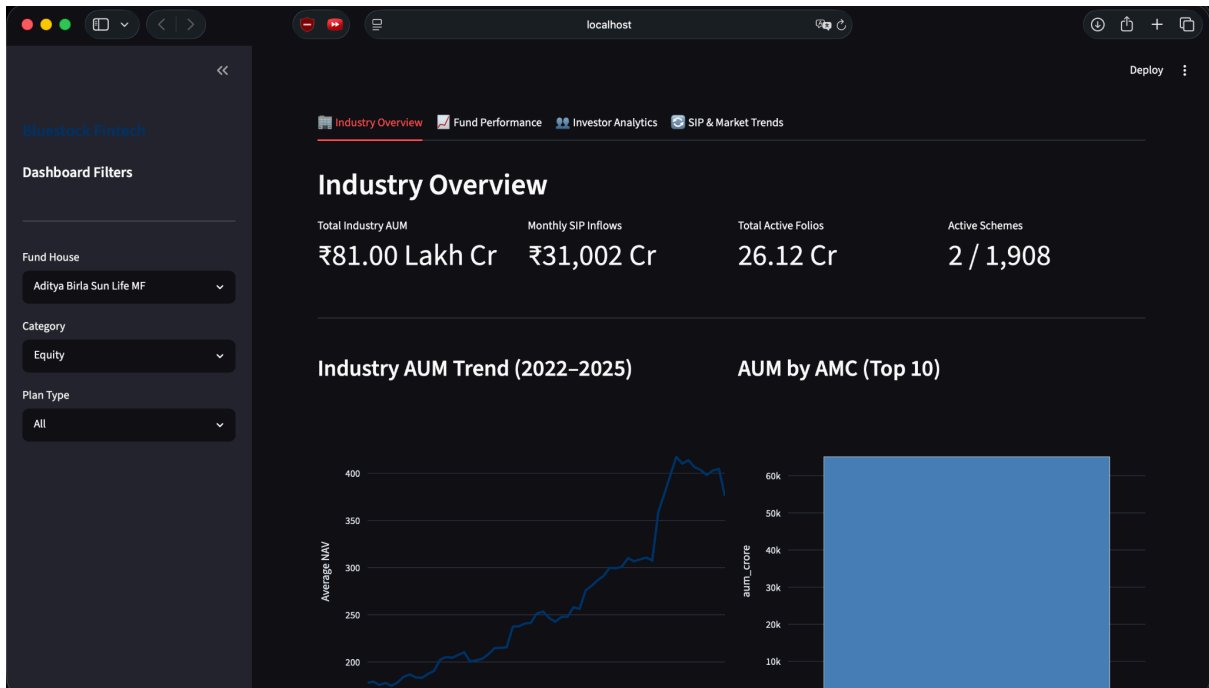


Figure 5: Streamlit Dashboard - Filters.

7. Investor Behavioral Analytics

A critical requirement of the platform was to transition from tracking funds to tracking human behavior, specifically focusing on customer retention.

7.1 Vintage Cohort Analysis

Investors were grouped based on their initial transaction year. The analysis revealed that cohorts onboarded during bear markets tend to maintain higher average monthly SIP amounts over time compared to cohorts that joined during speculative bull-market peaks.

7.2 SIP Continuity and Churn Prediction

By analyzing the time delta between recurring SIP transactions for accounts with a minimum of 6 historical orders, the system successfully identified workflow breakages. A total of 1,360 investor accounts were flagged as "at-risk" because their execution spacing gap exceeded 35 days, indicating mandate failures or intentional churn.

8. Algorithmic Recommender System

To demonstrate programmatic automation, a standalone Python module (recommender.py) was engineered.

This engine accepts a mock investor risk profile (Low, Moderate, or High) via the Command Line Interface. It dynamically queries the SQLite dimension tables, filters the universe of funds to match the exact risk tolerance, and sorts the results to return the Top 3 schemes ranked by their historical Sharpe Ratio. This lays the foundation for a robo-advisory feature.

9. Limitations & Future Scope

While the platform is highly functional, a few limitations exist:

- **Static Mock Data:** The investor demographics and transactional logs are limited to historical snapshots. Integrating a live bank API would be required for real-time production churn alerts.
- **Rolling Window Constraints:** The 90-day Sharpe ratio provides excellent short-term insight, but expanding the engine to calculate multi-year rolling metrics would require substantially more memory overhead.
- **Model Complexity:** Currently, the Recommender system matches purely on Risk Grade and Sharpe Ratio. Future iterations should incorporate Machine Learning (e.g., collaborative filtering) to recommend funds based on the behavior of similar demographic cohorts.

10. Strategic Business Recommendations

Based on the quantitative and behavioral insights generated by the Bluestock platform, the following strategic actions are recommended:

1. **Deploy Automated Churn Interventions:** Bluestock should immediately integrate the SIP Continuity tracking report with an automated email or SMS gateway. Engaging the 1,360 "at-risk" investors on day 32 of their payment gap could salvage significant recurring revenue.
2. **Introduce Tail-Risk Disclosures:** Given the steep CVaR drops observed in Sector funds, the UI should be updated to display explicit "Concentration Risk" warnings when retail investors attempt to allocate more than 30% of their portfolio to funds with an HHI greater than 2500.
3. **Target Tier-2 Cohorts:** Marketing spend should be reallocated to target Tier-2 and Tier-3 cities for SIP registrations. While Tier-1 cities drive lumpsum volume, the data proves that non-metro cohorts possess higher long-term retention and stickiness.

— End of Report —